

Isolation of Umbilical Cord-Derived Mesenchymal Stem Cells with High Yields and Low Damage.

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Mesenchymal stem cells (MSCs) are a population of multipotent cells with remarkable regenerative and immunomodulatory properties. Wharton's jelly (WJ) from the umbilical cord (UC) has gained increasing interest in the biomedical field as an outstanding source of MSCs. However, challenges such as limited supply and lack of standardization in existing methods have arisen. This article presents a novel method for enhancing MSC yield by dissecting intact WJ from the umbilical cord. The method employs blunt dissection to remove the epithelial layer, maintaining the integrity of the entire WJ and resulting in an increased quantity and viability of harvested MSCs. This approach significantly reduces WJ waste compared to conventional sharp dissection methods. To ensure the purity of WJ-MSCs and minimize external cellular influence, a procedure utilizing internal tension to peel off the endothelium after flipping the UC was conducted. Additionally, the Petri dish was inverted for a short time during explant culture to improve attachment and cell outgrowth. Comparative analysis demonstrated the superiority of the proposed method, showing a higher yield of WJ and WJ-MSCs with better viability than traditional methods. The similar morphology and expression pattern of cell surface markers in both methods confirm their characterization and purity for various applications. This method provides a high-yield and high-viability approach for WJ-MSC isolation, demonstrating great potential for the clinical application of MSCs.

Effects of MRI on stemness properties of Wharton's jelly-derived mesenchymal stem cells.

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Mesenchymal stem cells (MSCs), derived from various tissues, are served as a promising source of cells in clinic and regenerative medicine. Umbilical cord-Wharton's jelly (WJ-MSCs)-derived MSCs exhibit advantages over those from adult tissues, such as no ethical concerns, shorter population doubling time, broad differentiation potential, readily available non-invasive source, prolonged maintenance of stemness properties. The aim of this study was to evaluate the effect of MRI (1.5 T, 10 min) on stemness gene expression patterns (OCT-4, SOX-2, NANOG) of WJ-MSCs. Additionally, we assessed cell viability, growth kinetics and apoptosis of WJ-MSCs after MRI treatment. The quantitative real-time reverse transcription polymerase chain reaction (qRT-PCR) data showed that transcript levels of SOX-2, NANOG in MRI-treated WJ-MSCs were increased 32- and 213-fold, respectively. MTT assay was performed at 24, 48, and 72 h post-treatment and the viability was not significantly different between the two groups. The doubling time of the MRI group was markedly higher than the control group. In addition, the colony formation ability of WJ-MSCs after MRI treatment significantly increased. Furthermore, no change in apoptosis was seen before or after MRI treatment. Our results suggest that the use of MRI can improve the quality of MSCs and enhance the efficacy of mesenchymal stem cell-based therapies.

Therapeutic Potential in Wound Healing of Allogeneic Use of Equine Umbilical Cord Mesenchymal Stem Cells.

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Wound healing after skin injury is a complex process, particularly in equines where leg wounds are prevalent and their repair is complicated due to the anatomical characteristics. Conventional treatments are not effective enough. The umbilical cord offers an unlimited source of adult mesenchymal stem cells (ucMSCs) from Wharton's jelly tissue. The present study aims to demonstrate the safety and therapeutic potential of the allogeneic use of equine ucMSCs (e-ucMSCs) in the healing of severe equine leg wounds. The methods employed were the isolation, culture and expansion of e-ucMSCs. Flow cytometry and a PCR assay were used for cell characterization. This study included an immunomodulation assay, a murine pre-clinical trial and the first phase of an equine clinical trial. Our results showed that e-ucMSCs express a functional HLA-G homolog, EQMHCB2. In the immunomodulation assay, the e-ucMSCs inhibited the proliferation of activated equine peripheral blood mononuclear cells (e-PBMCs). In the murine pre-clinical trial, e-ucMSCs reduced healing time by 50%. In the equine clinical trial, the injection of e-ucMSCs into severe leg lesions improved the closure time and quality of the tissues involved, regenerating them without fibrous tissue scar formation. In conclusion, the results of this study suggest that e-ucMSCs can be used allogeneically for wound healing by creating a tolerogenic environment.

Induction of human umbilical Wharton's jelly-derived mesenchymal stem cells toward motor neuron-like cells.

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The most important property of stem cells from different sources is the capacity to differentiate into various cells and tissue types. However, problems including contamination, normal karyotype, and ethical issues cause many limitations in obtaining and using these cells from different sources. The cells in Wharton's jelly region of umbilical cord represent a pool source of primitive cells with properties of mesenchymal stem cells (MSCs). The aim of this study was to determine the potential of human Wharton's jelly-derived mesenchymal stem cells (WJMSCs) for differentiation to motor neuron cells. WJMSCs were induced to differentiate into motor neuron-like cells by using different signaling molecules and neurotrophic factors in vitro. Differentiated neurons were then characterized for expression of motor neuron markers including nestin, PAX6, NF-H, Islet 1, HB9, and choline acetyl transferase (ChAT) by quantitative reverse transcription PCR and immunocytochemistry. Our results showed that differentiated WJMSCs could significantly express motor neuron biomarkers in RNA and protein levels 15 d post induction. These results suggested that WJMSCs can differentiate to motor neuron-like cells and might provide a potential source in cell therapy for neurodegenerative disease.

New emerging potentials for human Wharton's jelly mesenchymal stem cells: immunological features and hepatocyte-like differentiative capacity.

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In recent years, human mesenchymal stem cells (MSC) have been extensively studied. Their key characteristics of long-term self-renewal and a capacity to differentiate into diverse mature tissues favor their use in regenerative medicine applications. Stem cells can be found in embryonic and extraembryonic tissues as well as in adult organs. Several reports indicate that cells of Wharton's jelly (WJ), the main component of umbilical cord extracellular matrix, are multipotent stem cells, expressing

markers of bone marrow mesenchymal stem cells (BM-MSC), and giving rise to different cellular types of both connective and nervous tissues. Wharton's jelly mesenchymal stem cells (WJ-MSC) express markers previously characterized in embryonic stem cells (ESC), such as Nanog and Oct3/4A. WJ-MSC further emerge as promising hypoimmunogenic cells, due to the expression of molecules able to modulate NK cells and expand regulatory T-cell populations. Moreover, it is now accepted that the differentiative capacities of such cells span all the mesoderm-derived tissues, extending to neuroectodermal as well as endodermal lineages. In this review, we compare very recent data on the potential of WJ-MSC to undergo hepatocyte-like differentiation with the results obtained from other adult MSC populations. Data in the literature strongly suggest that WJ-MSC can differentiate into diverse cell types, showing a unique ability to cross lineage borders. This, together with their in vitro proliferative potential and their immunoregulatory features, renders these cells extremely promising for regenerative medicine applications in different pathological settings.

Repair of Osteochondral Defects Using Human Umbilical Cord Wharton's Jelly-Derived Mesenchymal Stem Cells in a Rabbit Model.

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Umbilical cord Wharton's jelly-derived mesenchymal stem cell (WJMSC) is a new-found mesenchymal stem cell in recent years with multiple lineage potential. Due to its abundant resources, no damage procurement, and lower immunogenicity than other adult MSCs, WJMSC promises to be a good xenogenous cell candidate for tissue engineering. This in vivo pilot study explored the use of human umbilical cord Wharton's jelly mesenchymal stem cells (hWJMSCs) containing a tissue engineering construct xenotransplant in rabbits to repair full-thickness cartilage defects in the femoral patellar groove. We observed orderly spatial-temporal remodeling of hWJMSCs into cartilage tissues during repair over 16 months, with characteristic architectural features, including a hyaline-like neocartilage layer with good surface regularity, complete integration with adjacent host cartilage, and regenerated subchondral bone. No immune rejection was detected when xenograft hWJMSCs were implanted into rabbit cartilage defects. The repair results using hWJMSCs were superior to those of chondrogenically induced hWJMSCs after assessing gross appearance and histological grading scores. These preliminary results suggest that using novel undifferentiated hWJMSCs as seed cells might be a better approach than using transforming growth factor- β -induced differentiated hWJMSCs for in vivo tissue engineering treatment of cartilage defects. hWJMSC allografts may be promising for clinical applications.

The Effect of Human Mesenchymal Stem Cells Derived from Wharton's Jelly in Spinal Cord Injury Treatment Is Dose-Dependent and Can Be Facilitated by Repeated Application.

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Human mesenchymal stem cells derived from Wharton's jelly (WJ-MSCs) were used for the treatment of the ischemic-compression model of spinal cord injury in rats. To assess the effectivity of the treatment, different dosages (0.5 or 1.5 million cells) and repeated applications were compared. Cells or saline were applied intrathecally by lumbar puncture for one week only, or in three consecutive weeks after injury. Rats were assessed for locomotor skills (BBB, rotarod, flat beam) for 9 weeks. Spinal cord tissue was

morphometrically analyzed for axonal sprouting, sparing of gray and white matter and astrogliosis. Endogenous gene expression (Gfap, Casp3, Irf5, Cd86, Mrc1, Cd163) was studied with quantitative Real-time polymerase chain reaction (qRT PCR). Significant recovery of functional outcome was observed in all of the treated groups except for the single application of the lowest number of cells. Histochemical analyses revealed a gradually increasing effect of grafted cells, resulting in a significant increase in the number of GAP43+ fibers, a higher amount of spared gray matter and reduced astrogliosis. mRNA expression of macrophage markers and apoptosis was downregulated after the repeated application of 1.5 million cells. We conclude that the effect of hWJ-MSCs on spinal cord regeneration is dose-dependent and potentiated by repeated application.

Mesenchymal Stem Cell-Derived Exosomes Attenuate Hepatic Steatosis and Insulin Resistance in Diet-Induced Obese Mice by Activating the FGF21-Adiponectin Axis.

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Exosomes derived from mesenchymal stem cells have shown promise in treating metabolic disorders, yet their specific mechanisms remain largely unclear. This study investigates the protective effects of exosomes from human umbilical cord Wharton's jelly mesenchymal stem cells (hWJMSCs) against adiposity and insulin resistance in high-fat diet (HFD)-induced obese mice. HFD-fed mice treated with hWJMSC-derived exosomes demonstrated improved gut barrier integrity, which restored immune balance in the liver and adipose tissues by reducing macrophage infiltration and pro-inflammatory cytokine expression. Furthermore, these exosomes normalized lipid metabolism including lipid oxidation and lipogenesis, which alleviate lipotoxicity-induced endoplasmic reticulum (ER) stress, thereby decreasing fat accumulation and chronic tissue inflammation in hepatic and adipose tissues. Notably, hWJMSC-derived exosomes also promoted browning and thermogenic capacity of adipose tissues, which was linked to reduced fibroblast growth factor 21 (FGF21) resistance and increased adiponectin production. This process activated the AMPK-SIRT1-PGC-1 α pathway, highlighting the role of the FGF21-adiponectin axis. Our findings elucidate the molecular mechanisms through which hWJMSC-derived exosomes counteract HFD-induced metabolic dysfunctions, supporting their potential as therapeutic agents for metabolic disorders.

Are serum-free and xeno-free culture conditions ideal for large scale clinical grade expansion of Wharton's jelly derived mesenchymal stem cells? A comparative study.

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Mesenchymal stromal/stem cells (MSCs) for clinical use have largely been isolated from the bone marrow, although isolation of these cells from many different adult and fetal tissues has been reported as well. One such source of MSCs is the Whartons Jelly (WJ) of the umbilical cord, as it provides an inexhaustible source of stem cells for potential therapeutic use. Isolation of MSCs from the umbilical cord also presents little, if any, ethical concerns, and the process of obtaining the cord tissue is relatively simple with appropriate consent from the donor. However, a great majority of studies rely on the use of bovine serum containing medium for isolation and expansion of these cells, and porcine derived trypsin for dissociating the cells during passages, which may pose potential risks for using these cells in clinical applications. It is therefore of high priority to develop a robust production process by optimizing culture variables to

efficiently and consistently generate MSCs that retain desired regenerative and differentiation properties while minimizing risk of disease transmission. We have established a complete xeno-free, serum-free culture condition for isolation, expansion and characterization of WJ-MSCs, to eliminate the use of animal components right from initiation of explant culture to clinical scale expansion and cryopreservation. Growth kinetics, in vitro differentiation capacities, immunosuppressive potential and immunophenotypic characterization of the cells expanded in serum-free media have been compared against those cultured under standard fetal bovine serum (FBS) containing medium. We have also compared the colony-forming frequency and genomic stability of the large scale expanded cells. Secretome analysis was performed to compare the angiogenic cytokines and functional angiogenic potency was proved by Matrigel assays. Results presented in this report identify one such serum-free, xeno-free medium for WJ expansion. Cells cultured in serum-free, xeno-free medium exhibit superior growth kinetics and functional angiogenesis, alongside other MSC characteristics. We report here that WJ-MSCs cultured and expanded in Mesencult XF, SF Medium retain all necessary characteristics attributed to MSC for potential therapeutic use.

Tailoring cell therapies for diabetic metabolic phenotypes: a comparative study on the efficacy of various umbilical cord-derived cell regimens.

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Given the high heterogeneity of type 2 diabetes mellitus (T2DM), it is imperative to develop personalized stem cell infusion regimen for targeted metabolic phenotype in order to ensure optimal therapeutic efficacy. In this study, we conducted a comparative analysis of 4 infusion regimens involving single and repeated infusions of human umbilical cord Wharton's jelly-derived MSCs (hucMSCs), single infusions of umbilical cord blood mononuclear cells (UCB), and sequential infusions of hucMSCs and UCB in T2DM rats. Results showed all 4 infusion regimens exhibited comparable efficacy in lowering fasting blood glucose levels and suppressing glucagon secretion. Single and double infusions of hucMSCs exhibited a tendency to migrate to the liver, thereby better at ameliorating hepatic glucose metabolism by enhancing glycogen synthesis and storage, promoting glycolysis, inhibiting gluconeogenesis, and improving insulin signal transduction. The sequential infusion of hucMSCs and UCB demonstrated specific cell tropism toward the pancreas, leading to prolonged glucose-lowering effects following a glucose tolerance test, restoration of early-phase insulin secretion, stimulation of islet beta cell proliferation and improvement in the beta/alpha ratio. Multiple injections, regardless of cell type, reduced the expression of systemic chronic inflammatory markers such as IL-1 β , IL-6, IL-17, IL-22, and IFN- γ . Finally, a single dose of UCB exhibited a greater tendency to target visceral fat and enhanced effectiveness in regulating levels of total cholesterol and triglycerides. In conclusion, our study provided personalized stem cell regimens for diverse T2DM metabolic phenotypes, thereby offering improved treatment alternatives for future clinical trials and applications.